Suggested Readings and Repos

Readings

- There Will Be a Scientific Theory of Deep Learning
  https://arxiv.org/abs/2604.21691
- Training Compute-Optimal Large Language Models
  https://arxiv.org/abs/2203.15556
- Muon is Scalable for LLM Training
  https://arxiv.org/abs/2502.16982

Open source repos

- karpathy/nanochat: full training pipeline for GPT-2 style model
- kellerjordan/modded-nanogpt: lowest time given a constrained loss $L(T)$
- openai/parameter-golf: lowest loss given fixed parameters $L(N)$
- qlabs-eng/slowrun: lowest loss given constrained data and infinite compute $L(D)$
- qlabs-eng/noise: pretraining without backpropagation (lowest loss or training time)
- kellerjordan/modded-nanogpt: lowest step count given infinite time (optimizer track)